Abdul Khayyum Farooqui
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EDUCATION

Rutgers University, New Brunswick, NJ, USA
Master of Science, Statistics – Data Science
Sreenidhi Institute of Science and Technology, Hyderabad, India
Bachelor of Technology, Computer Science and Engineering

TECHNICAL AND OTHER SKILLS

Analytic Tools and Languages: Python, R, PySpark, SQL, NoSQL, Matlab, Advanced Microsoft Excel, GitHub, JIRA, Databricks, Apache Spark, Hadoop
Database and Visualization: SQL, Snowflake, Oracle DB, Power BI, Tableau, Excel, Matplotlib, Splunk
Data Science and Analytics: A/B Testing, Machine Learning, Deep Learning, Natural Language Processing (NLP), Predictive Modeling, Statistical Analysis, Statistical Modeling, Clustering, K-means, Regression, Random Forest, XG-Boost, Embedding, TensorFlow, Keras, Scikit-learn, pandas, Airflow, Spark, Big Data Analytics, Data Warehousing, ETL Processes

PROFESSIONAL EXPERIENCE

Python Developer Intern, *Altech Power and Energy Systems Private Limited* *Feb 2023—May 2023*
- Developed an NLP-based next-word prediction model using Python and TensorFlow, optimizing computational efficiency by 12.7% and reducing memory usage by 6.8% through hyperparameter tuning
- Managed a 3-person data science team to design and deploy two predictive models, leveraging Python and R for comparative analysis, improving model speed by 6% and reducing memory consumption by 17%

Data Visualization Intern, *Globalshala* *Jun 2022—July 2022*
- Led a 6-member analytics team to build ETL workflows and analyze digital ad performance metrics (CTR, impressions, ROI) using SQL and Python, improving data accuracy by 20% and enabling KPI-driven campaign optimization.
- Conducted ROI analysis on ad campaigns using SQL and Tableau, identifying underperforming ads and implementing data-driven optimizations that reduced advertising costs by 15%

Data Science Intern, *RemarkSkill IIT* *May 2022—Jun 2022*
- Applied machine learning algorithms to classify customers based on shopping patterns, improving segmentation accuracy by 22%
- Engineered a customer segmentation model using K-Means clustering and DBSCAN, optimizing marketing strategies by categorizing customers based on purchasing frequency and product preferences, leading to a 15% increase in targeted marketing efficiency

ACADEMIC PROJECTS

Tesla Stock Price Prediction and Sentiment Analysis
- Built a forecasting pipeline to predict Tesla's next-day closing price using Random Forest and XGBoost with quantile regression, incorporating MACD, RSI, Bollinger Bands and FinBERT-derived sentiment features from Reddit, The Guardian, and NYT
- Implemented dynamic Plotly visualizations to examine feature contributions, confidence intervals, and relationships between sentiment trajectories and price volatility, supporting model interpretability and exploratory analysis

Breast Cancer Classification
- Developed a machine learning pipeline for breast cancer classification using logistic regression and support vector machines (SVM), achieving 91.11% accuracy, 94.34% sensitivity, and 86.49% specificity in detecting malignant tumors
- Engineered a feature extraction framework combining morphological and texture-based attributes, optimizing predictive performance and reducing classification error by 15% in ultrasound image analysis

PUBLICATIONS

Deep Reinforcement Learning for Autonomous Robot Navigation in Dynamic Environments - *Institute of Electrical and Electronics Engineers (IEEE)*
- Built a DRL model with TRPO, increasing accuracy 30%, cutting overhead 20%, and improving efficiency 25%
- Developed a CNN-based vision system, boosting detection 27%, reducing false positives 32%, and speeding processing 18%

Machine Learning based Big Data applications for E-Commerce Logistics - *International Journal of Engineering, Science, Technology and Innovation (IJESTI)*
- Built a machine learning-based customer segmentation model using CNNs and gradient boosting, boosting targeted marketing efficiency by 20% and increasing purchase conversions by 15%
- Mapped and analyzed supply chain process data (order cycle, inventory turnover) to define performance metrics and improve end-to-end visibility using automated data pipelines.

Rides Request Demand Forecast – OLA Bike - *International Research Journal of Engineering and Technology (IRJET)*
- Applied clustering to segment ride zones and modeled demand-supply gaps, contributing to process improvements in real-time ride allocation and fleet optimization.
- Optimized real-time ride allocation by 22% using Mini-Batch K-Means, clustering 4M data points into 50 zones

Hybrid Reinforcement Learning based GA for Portfolio Optimization - *Journal of Northeastern University*
- Developed a Q-learning GA for portfolio optimization, increasing returns by 22%, reducing risk by 18%, and cutting error by 49.7%
- Enhanced investment strategy efficiency with 10% higher stability, 27% faster convergence, and 15% lower computational costs